

Machine Learning–Based Prediction of Removal Efficiency Using Regime-Aware, Physics-Informed Gradient Boosting

1. Introduction

Removal efficiency (RE) is a critical metric in adsorption-based water treatment systems. Experimental adsorption data collected from the literature exhibit substantial heterogeneity due to variations in adsorbents, adsorbates, operating conditions, and experimental protocols. To address these challenges, this work develops a regime-aware, physics-informed machine learning framework.

2. Dataset and Target Variable

The dataset consists of experimentally reported adsorption studies compiled from literature. The target variable is removal efficiency (RE, %). Input features include surface area, adsorption capacity, initial concentration, contact time, adsorbent dose, stirring speed, pH, temperature, and adsorbate identity.

3. Data Cleaning and Preprocessing

Literature-reported numeric values often contain inconsistencies such as decimal commas, uncertainty notation, and approximate values. A robust numeric cleaning procedure was applied to standardize all inputs. Rows with missing or physically invalid values were removed, and RE was constrained to the range 0–100%.

4. Physics-Informed Feature Engineering

Derived features were introduced to embed adsorption physics into the model, including site density, capacity-to-loading ratio, logarithmic contact time, mixing index, and thermo-capacity coupling. These features improved both model performance and interpretability.

5. Regime Definition

To address non-stationarity, the dataset was divided into regimes based on solution pH and initial concentration. Acidic and neutral conditions were separated using $\text{pH} = 7$, while low and high concentration regimes were defined relative to the median initial concentration.

6. Restricted-Domain Modeling

Regimes with fewer than 10 samples were excluded to avoid unstable generalization. This restricted-domain approach reduced noise while preserving physical relevance.

7. Model Development and Comparison

Multiple models were evaluated, including linear regression, random forest, SVR, XGBoost, and LightGBM. Tree-based gradient boosting methods performed best due to their ability to capture nonlinear interactions.

8. XGBoost Benchmark

A regime-aware XGBoost model with sample weighting and residual learning achieved a test R^2 of 0.809 and RMSE of 13.44, establishing a strong baseline.

9. LightGBM Final Model

LightGBM, using leaf-wise tree growth with strong regularization, achieved the highest performance with a test R^2 of 0.847 and RMSE of 12.05. This improvement is attributed to better capture of localized nonlinearities.

10. Model Diagnostics and Interpretability

Residual analysis indicated unbiased predictions with no significant heteroscedasticity. SHAP analysis identified temperature, capacity-to-loading ratio, adsorption capacity, and pH as the most influential features, consistent with adsorption theory.

11. Limitations

The alkaline regime could not be modeled due to insufficient data. The dataset remains heterogeneous, and predictions should not be extrapolated beyond observed regimes.

12. Conclusion

A physics-informed, regime-aware LightGBM model was successfully developed to predict adsorption removal efficiency. The framework combines strong predictive accuracy with physical interpretability, achieving an R^2 of 0.847, and is suitable for practical screening and analysis of adsorption systems.